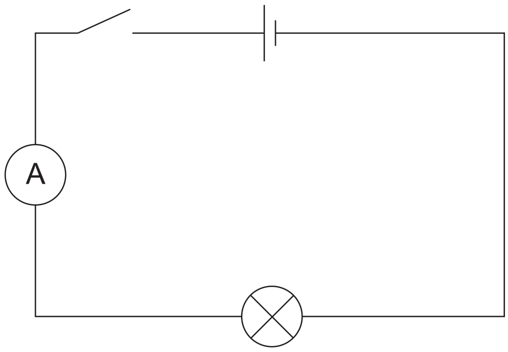
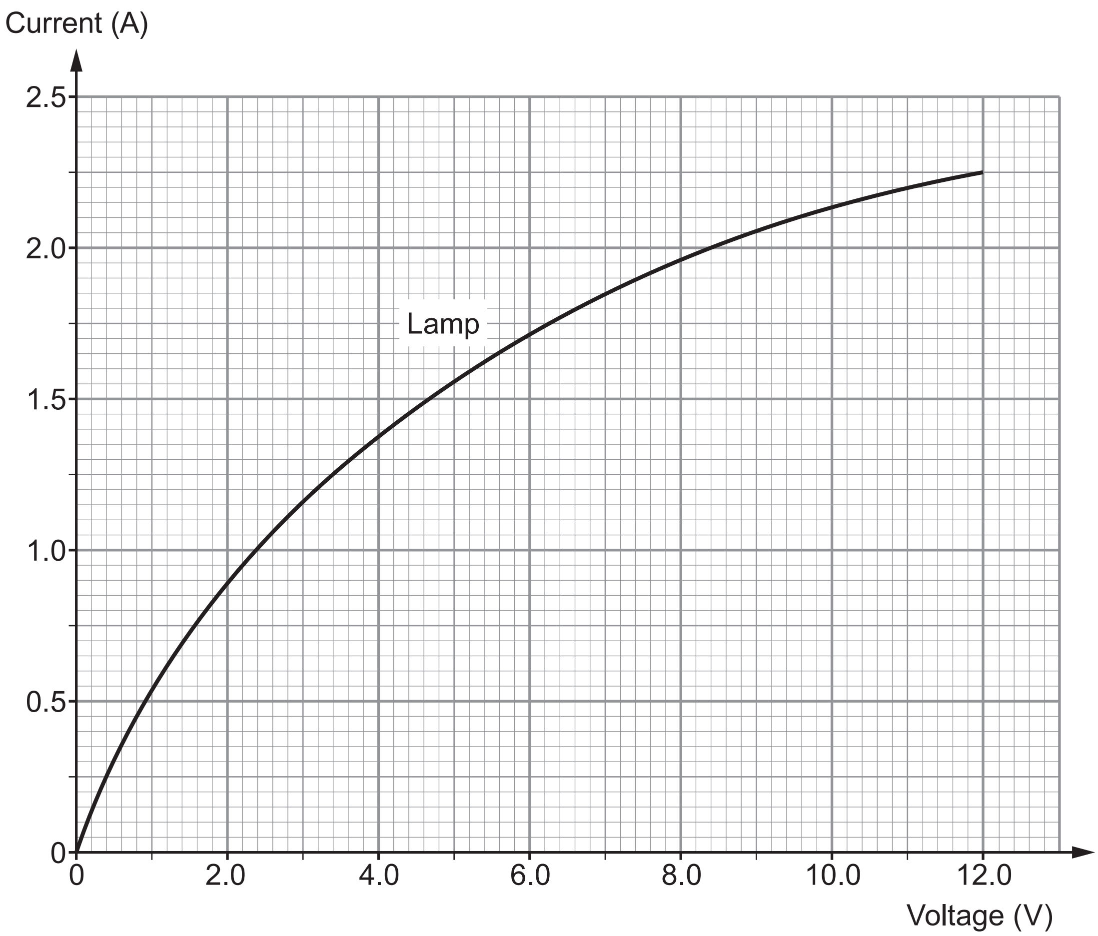
Examiner only

6. A group of pupils investigate the current-voltage (I-V) characteristics of different components.

(a) The first component they investigate is a lamp.
Part of the circuit used is shown below.
Add a variable resistor **and** voltmeter to the circuit diagram. [2]



(b) They draw a graph from their results for the **lamp**. It is shown below.



Examiner only

(i) One student suggests that as the current through the lamp **doubles** the voltage **triples**.
Use pairs of data within the range 0.5 A to 2.0 A from the graph to explain whether you agree with the student. [3]

....

....

(ii) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

and information from the graph to calculate the **maximum** power produced by the lamp. [3]

Power = ... W

(c) The experiment is repeated with a 6 Ω resistor but the results are lost.

(i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

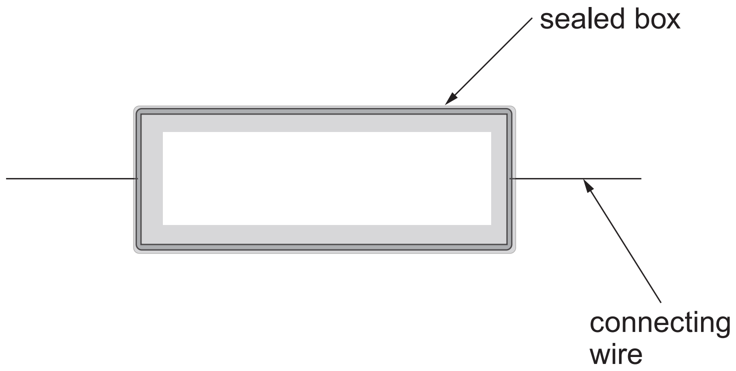
to calculate the current through the 6 Ω resistor at 12 V. [2]

Current = ... A

(ii) **Draw the line for this resistor** on the grid on the previous page. [1]

Examiner only

(d) The students are given a sealed box containing another component.



They are asked to confirm whether the hidden component is a diode.

(i) Describe how they would use the circuit shown in part (a) to confirm whether or not it is a diode. [3]

....

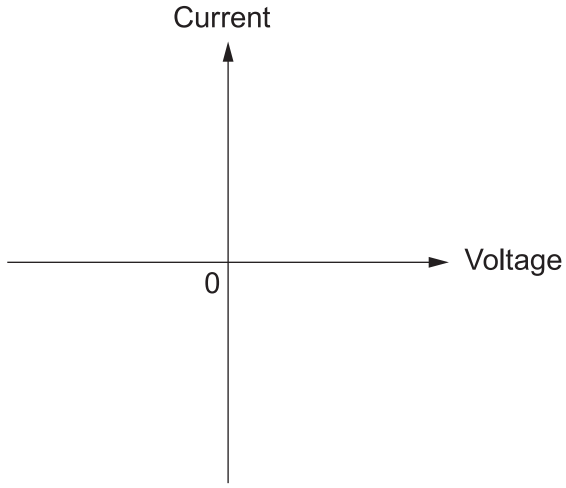
....

....

....

....

(ii) **Sketch** the I-V graph that you would expect to obtain for a diode on the grid below. [1]



15

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